<u>**Software Tools 2025 - Assignment #4 Instructions – File 1 of 2: Candidate Projects**</u>

See also file 2 of 2, with the detailed project preparations instructions and grading rubric.

<u>**Overview of Assignment #5 Scope and General Tips**</u>

Your final project for Software Tools is an opportunity to conduct a small yet meaningful project of interest to you, while practicing your skills in data handling and using various software tools. The Assignment #4 scope actually should be larger than Assignments #1 in terms of length or time commitment. But remember, please focus on <u>**quality.**</u> You will put into use a variety of programming skills as well as critical thinking skills for data quality assessment and analysis.

After you complete an end-to-end project draft, I suggest going through your code again to see what improvements you can make to your project. For example, can you reduce code redundancy through using an apply function or writing your own function? Can you improve your visualizations?

You are welcome to choose a project topic that will help you with your future work, such as your BINF*6999 project or your thesis. Please note that "double dipping" isn't permitted for academic credit. This means that you can't copy/paste an assignment from one course to another course or directly into your MSc or PhD thesis. However, there are many ways you can structure this assignment to be useful for you. For example, using publicly available data or data from another lab member, you could try out one or more software tools that you are considering using for your future research. On the other hand, some students may wish to choose a different topic from your thesis in order to help you to gain a wider variety of analytical experiences. If you are in doubt, don't hesitate to speak with me to ensure that your project is suitable for credit here.

Below, I have provided several example project ideas. You are encouraged to consider choosing from among these, as each involves building in a logical and feasible way upon topics or skills we have covered in class to varying degrees. <u>**If you choose from among these, then there is no need to seek project topic approval.**</u>

You are also welcome to choose your own topic. This project is intended to be completed through regular, but part-time, effort over the duration of 3-4 weeks. Therefore, the scope should be kept manageable. <u>**If you are choosing your own topic, then you should speak with me at least three weeks before the due date.**</u>

The data set sizes are guidelines only, not hard rules. You may go somewhat above these sample sizes if you wish to do so, and if suitable for your project. Your project should run readily on a desktop or laptop computer.

The majority of your script should be written in R, but, if suitable for your project, you may incorporate an additional software tool (e.g. RAxML, MEGA, PICRUSt2, etc.). If you choose to do so, then you need to explain very clearly what you did! Provide all scripts and analysis files.

Make sure that all input files needed are provided so that your R script will run end to end; e.g. if one of your inputs to R is an output from another software, then provide that file.

Before selecting your topic, I encourage you to scroll through all of the recommended links, so that you are aware of the existence of these various analytical tools and visualization methods. And, we hope you have fun!

Recommended Timeline for Success

It is vital to start this assignment early and to work on this regularly over the duration of 3-4 weeks. Here is an example timeline that will help you to be as successful as possible and avoid stress at the end.

Week 1 – Read over the candidate projects and assignment preparation instructions carefully. Bring questions to class. Choose 2 themes of interest. Explore the provided vignettes and other tutorials of interest to you.

Week 2 – Choose your specific project objective. Obtain and explore 1-2 candidate data sets. Consider sample size and suitability of the data for your objective, and conduct data exploration and exploratory plotting. Read several papers, and prepare a point-form outline of your introduction.

Week 3 – Complete a draft of your commented code and write out your introduction. Complete at least point-form notes for all of the other written sections.

Week 4. Complete an end-to-end draft early in the week. Refine your assignment. Look at the grading rubric again, and aim to move up a level, if suitable. Run your code in a new RStudio session to ensure it runs. Proofread your work. Submit early.

Topics for Assignment #4

Candidate Project Themes

1) Supervised Machine Learning and Classification Using Sequence Features

- Using two or more genes from NCBI (or other public sequence data base) for a single taxonomic group, build and test a classifier for separating these genes. Alternatively, you could build a taxonomic classifier. As part of your project, you may adapt the randomForest example script from class. Be sure to consider all steps and arguments to functions used; don't just copy/paste: adapt the script as suitable for your specific problem and data set. If you do use the randomForest example script as a key resource, then you must add at least one more classification method for comparison with random forest. In the written portions, show that you understand what the methods are doing, not only that you can run them. Be sure to test your model on validation data (unseen during model training). You may also consider using cross-validation for your project. As well, be sure to include at least one visualization suitable for a machine learning project for presenting your results, e.g. receiver operating characteristic (ROC) curve.

- This project would be a great choice for those interested in going into more depth in understanding and applying machine learning algorithms beyond the course materials.
- Tip: If your classification problem ends up being easy to solve (e.g. if you easily get 100% classification success on your training and validation data with simple sequence features), then I suggest to report that outcome, but you may also wish to try out a second data set or problem that could pose a more interesting and challenging classification problem (e.g. move from phylum or class to genus-level or species-level classification problem). The second problem should be connected to the first problem, for a cohesive final document.
- Tips: I suggest using a modest-sized dataset for this theme (e.g. 1,000-10,000 sequences total if you will use features such as k-mer frequencies). Consider trying various k-mer lengths to assess impact upon your classifier.
- Some helpful links relating to this theme:
- https://www.kaggle.com/camnugent/introduction-to-machine-learning-in-r-tutorial
- https://www.datacamp.com/community/tutorials/machine-learning-in-r
- https://lgatto.github.io/IntroMachineLearningWithR/an-introduction-to-machine-learning-with-r.html
- https://www.machinelearningplus.com/machine-learning/caret-package/

2) Unsupervised Machine Learning or Clustering of Sequences

- Do different genes exhibit different clustering patterns within the same taxonomic groups? Such a project has relevance for determining if different genes are more or less suitable as markers for future classification applications and also can reveal differences in evolutionary pressures (e.g. functional constraint) among genes.
- This project would be a great choice for students interested in machine learning, genetics, and biodiversity.
- Steps in this project: For a taxonomic group of interest (I suggest to choose a genus or possibly a family), obtain DNA sequence data for two different genes from NCBI or another data base. Using either extracted sequence features (e.g. k-mer frequencies) or alignments (if the dataset size is modest), calculate a pairwise distance matrix among sequences for each gene. Then, cluster your sequences for each gene. If you wish, you may build upon our example script performing clustering of sequences using functions available through the DECIPHER package. It would then be required to expand upon the example scripts from class, such as by evaluating and comparing cluster strength using internal measures (e.g. Dunn index or Silhouette index). You could consider including a Silhouette plot as one of your visualizations. You could also consider using multivariate techniques (e.g. PCA, NMDS) to visualize your data.
- Tips: Your sample size of sequences should be relatively similar between genes. If one gene has a lot more data than the other, then you may consider eliminating highly similar

sequences for the gene with more data. You may also consider including a relatively similar number of species between the two genes.

- Tip: I suggest using a fairly modest dataset size for the scope of this assignment (e.g. 100-1000 sequences if you wish to use alignment-based methods, up to ~10,000 for a k-mer based analysis).
- Helpful links:
- https://lgatto.github.io/IntroMachineLearningWithR/unsupervised-learning.html
- https://cran.r-project.org/web/packages/kmer/kmer.pdf
- https://www.bioconductor.org/packages/release/bioc/html/DECIPHER.html
- https://cran.r-project.org/web/packages/cluster/cluster.pdf
- https://cran.r-project.org/web/packages/clValid/vignettes/clValid.pdf

3) Geography and Evolutionary Diversification

- How does geography relate to diversification? Does speciation within a given taxonomic group tend to occur within geographic regions, or does large-scale allopatric divergence contribute to evolutionary diversification? An example specific question under this theme would be: Do sister species for my taxon of choice live in the same geographic region or different geographic regions?
- This project would be a great choice for students interested in biogeography, phylogeny, geographic mapping, and cool visualizations.
- To address this question, you would: obtain DNA sequence data for a chosen gene from NCBI or another data base for your taxonomic group of choice, perform sequence alignment and phylogeny generation, obtain distribution information for the species in the genetic dataset (e.g. from GBIF and/or BOLD), match up the sequence and geographic data (e.g. using species name – mainly relevant for vertebrates due to problems of cryptic species in invertebrates), make a map, and compare the phylogeny and geography for the species in your data set (see links below). The beginnings of the phylogeny portion of your script may be adapted from class example scripts if you wish. If suitable for your data set, use a different alignment method (such as a secondary structure model for ribosomal RNA genes) or a codon-based alignment for protein-coding nucleotide sequences (e.g. see DECIPHER package functions). For your phylogeny construction, aim to go beyond the class materials, such as through conducting model testing (e.g. using the phangorn package to select the most suitable model of molecular evolution) and/or maximum likelihood phylogenetic analysis to reconstruct your tree. The geography/mapping portion would also be a novel component compared to class. There are several helpful tutorials in the links below for the mapping part.
- Tip: use a genus as your taxonomic level of focus; include all available species in that genus.
- Tip: use a small dataset (e.g. 10-100 sequences, after all data filtering steps). If your taxonomic group has a lot of data for your gene, you can select sequences for inclusion based upon, for example, discarding similar sequences (e.g. those that are <1% different, etc.). Or, you could trim sequences for the tree (e.g. one tip per species) but visualize all the geographic data points.

- Tip: use a gene that has support in the literature as having good signal for reconstructing phylogenetic relationships among species within genera (e.g. COI, cytb,12S, 16S, 28S). Considering the intended scope of this assignment, I suggest choosing a single gene, based upon good data availability for your target taxonomic group. You may also consider constructing a multi-gene tree if you wish to gain experience with that as a novel component of your assignment.
- Example helpful links:
- https://www.molecularecologist.com/2014/11/geophylogeny-plots-in-r-for-dummies/
- https://www.molecularecologist.com/2017/02/phylogenetic-trees-in-r-using-ggtree/
- https://cran.r-project.org/web/packages/phytools/phytools.pdf
- https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1600-0587.2010.06312.x
- http://blog.phytools.org/2019/03/projecting-phylogeny-onto-geographic.html
- http://www.phytools.org/Cordoba2017/ex/15/Plotting-methods.html

4) Trait Evolution

- Is a given trait phylogenetically conserved? (i.e. Do closely related species have similar trait values?) Mapping patterns in trait evolution on a phylogeny helps us to understand functional constraints, cases of rapid evolution, and convergence through evolutionary history.
- This project would be a great choice for students interested in molecular phylogenetics, evolutionary biology, and cool visualizations. This project can also be adapted to have a conservation angle. For example: Do evolutionarily similar species have similar or different conservation status (e.g. IUCN Red List status)? Are we at risk of losing major branches of the tree of life, representing unique evolutionary history?
- This project would involve: choose a specific taxonomic group with trait data available (see pre-recorded lecture on sources of trait data), obtain genetic data from NCBI or another database for this taxon, match up the sequence and trait data (e.g. based upon the species name), and then test for phylogenetic conservatism by estimating the lambda parameter (for a quantitative variable, such as body size). If you wish, you may adapt some portions of the phylogeny part from example class scripts. If suitable for your data set, use a different alignment method (such as a secondary structure model for ribosomal RNA genes) or a codon-based alignment for protein-coding nucleotide sequences (e.g. see DECIPHER package functions). For your phylogeny, aim to go beyond the class materials, such as through conducting model testing (e.g. using the phangorn package) and/or maximum likelihood phylogenetic analysis. The trait/sequence matching and lambda calculation would be a novel component. The associated main visualization would be to reconstruct the evolutionary history of the trait onto a phylogeny, for example using tools available in the R packages ape, phangorn, and phytools (see links below). I also suggest that you may consider making a visualization using the ggtree package that shows the distribution of trait values or the evolutionary history mapped onto the phylogeny.
- An alternative project under this theme is to see whether two traits (or a trait and environmental variable) are correlated with one another among species, taking into account

phylogenetic relationships (e.g. through phylogenetic generalized least squares (PGLS) analysis). E.g. Are latitude and body size correlated? Are body size and litter size correlated?

- Tip: use a small-sized dataset for this project (for example 20-100 species).
- Tip: use a gene that has support in the literature as having a good signal for reconstructing phylogenetic relationships among species in your taxonomic group. Common marker choices for reconstructing evolutionary relationships among species within genera would include: COI, cytb, 12S, 16S, and 28S (28S also used for deeper phylogenetics). You may use a single gene for your analysis, or you may consider building a multi-gene phylogeny as a novel component of your assignment.
- Tip: If you are searching for a gene such as 28S through NCBI, also include the long name in your search, e.g. "28S OR large ribosomal subunit". For 18S, which is often used for reconstructing deep evolutionary relationships, this would be: "18S OR small ribosomal subunit".
- Helpful links:
- http://www.phytools.org/static.help/phylosig.html
- http://www.phytools.org/Cordoba2017/ex/2/Intro-to-phylogenies.html
- http://www.phytools.org/Cordoba2017/ex/15/Plotting-methods.html
- https://4va.github.io/biodatasci/r-ggtree.html
- https://guangchuangyu.github.io/ggtree-book/chapter-ggtree.html
- http://www.phytools.org/Cordoba2017/ex/3/PICs.html

5) Molecular Phylogenetics and Congruence

- Do two genes yield the same or different phylogenetic hypotheses? Why or why not? We may get differences in resolution if there are differences in molecular evolutionary rates between genes, for example. Discordance between nuclear gene and mitochondrial gene trees can also be an indicator of a history of reticulate evolution (hybridization and introgression).
- This project would be a great choice for students interested in genetics, molecular phylogenetics, and visualization.
- For this question, you would choose a taxonomic group (of modest size) and obtain genetic data for two different markers. After determining which species have data for both genes, you would perform a sequence alignment for each gene and build a phylogenetic tree for each gene. If you wish, you may adapt class example scripts for some portions of the phylogenetics analysis portion. If suitable for your data set, use a different alignment method (such as a secondary structure model for ribosomal RNA genes) or a codon-based alignment for protein-coding nucleotide sequences (e.g. see DECIPHER package functions). For your phylogeny, aim to go beyond the class materials, such as through conducting model testing (e.g. using the phangorn package) and/or maximum likelihood phylogenetic analysis. Then, building upon that, you would examine topological congruence of the two trees through visualization, such as using functions available through the phytools, phangorn, or dendextend package or other resources. You could also consider calculating one or more formal metrics of phylogenetic congruence.

- Tip: Choose a small data set size for this project (e.g. 10-50 species for each of two genes, after data filtering steps).
- Tips: Consider performing this project for a genus or family-level taxonomic group. Choose genes that are supported as being helpful for resolving relationships among species within genera or families. Examples of common genes used for phylogenetics include COI, cytb for vertebrates, 12S, 16S, and 28S. You would choose two genes for this project.
- Helpful links:
- http://www.phytools.org/Cordoba2017/ex/15/Plotting-methods.html
- http://blog.phytools.org/2013/09/plotting-facing-trees-using-phytools.html
- https://cran.r-project.org/web/packages/phangorn/vignettes/IntertwiningTreesAndNetworks.html
- https://cran.r-project.org/web/packages/dendextend/vignettes/dendextend.html
- https://www.r-bloggers.com/dendextend-a-package-for-visualizing-adjusting-and-comparing-dendrograms-based-on-a-paper-from-bioinformatics/

6) Phylogenetic Community Structure

- The evolutionary relatedness of communities of organisms can provide us with insights into mechanisms of community formation, particularly environmental filtering. Under harsh environmental conditions, often closely related species (which share certain traits) are found co-existing, while communities in other geographic areas or environments are more phylogenetically diverse. Are co-occurring species more or less similar to one another than expected by chance?
- This project would be an excellent choice for students interested in molecular ecology, community ecology, biodiversity, phylogenetics, evolutionary biology, and/or microbiome.
- This project would involve the following steps. Choose a geographic region and taxonomic group of interest. As suitable for your taxonomic group, define "communities" within your broader region. For example, for aquatic organisms, communities could be lakes, rivers, or watersheds. Or, communities could be defined based upon spatial scale (e.g. a grid system at a scale suitable for your taxonomic group). For animal microbiome samples, a community could be those species inhabiting single host organisms. Obtain genetic data from NCBI or other public databases from a relevant marker for your group for building a phylogeny. Build a phylogenetic hypothesis for your full geographic region (i.e. full species pool). Build a community matrix, showing the site-by-taxon occupancy for each site or region. For some taxonomic groups, species would be a suitable choice for taxonomic level. For others, such as many invertebrates, BINs or genetic clusters could be more suitable. For 16S prokaryote data, you may consider whether you wish to use sequence clusters or ASVs (amplicon sequence variants) as your taxa. Then, conduct a test for whether cohabiting species are more phylogenetically related or less phylogenetically related than expected by chance. A suitable visualization could be a phylogenetic tree showing those tips forming one community (e.g.

via colour or symbol). This would help to provide a visualization to convey whether cohabiting species are phylogenetically clustered or not.

- Tip: use a modest-sized data set for this project (such as 50-1000 species, after data filtering steps)
- Tip: For this project, I suggest that you may consider using a single gene to build your phylogeny, to leave time for the next analytical steps involved.
- Helpful links:
- https://pedrohbraga.github.io/CommunityPhylogenetics-Workshop/CommunityPhylogenetics-Workshop.html
- https://cran.r-project.org/web/packages/picante/vignettes/picante-intro.pdf
- https://guangchuangyu.github.io/ggtree-book/chapter-ggtree.html
- https://vaulot.github.io/tutorials/Phyloseq_tutorial.html

7) Methods for Differential Gene Expression Analysis

- Gene expression analysis is an important and fascinating area of research, and it is important to be aware of the consequences of methodological choices upon conclusions. Are findings regarding differential gene expression robust and likely to be reproducible with further biological experiments? This project involves comparing and contrasting two or more methodological choices.
- This option would be an excellent project choice for students interested in gene expression analysis, benchmarking, statistics, and software evaluation.
- Details: For this project, first read Tong et al. (2020). Choose one of the most important methodological choices mentioned in their paper. Your choice of a step to focus on could be at any point in the analytical pipeline. If you want to look at a question relating to read mapping or counting algorithm, then you would need to obtain raw transcript sequence data set to analyze (e.g. from NCBI's Sequence Read Archive, SRA). Or, if you wish to look at the consequences of an analytical choice downstream of read mapping (such as normalization method – a very important choice), then you may obtain a gene expression dataset (e.g. from the Gene Expression Omnibus, GEO, of NCBI or from a specific paper or tutorial). Explore the impact of two different methodological choices upon the results of your analysis to detect genes with differential expression between samples. This could be using two different software tools, or this could involve choosing two or more very different options or parameter settings within one tool. How do the results differ with this choice? Choose at least two of the following criteria for comparing the results: computational speed, accuracy, precision, and reliability (see Tong et al. 2020).
- Tip: You may choose a data set that is provided with one or more of the Bioconductor vignettes if you wish, or you may obtain your own data set of interest from a publication or GEO.

- Tip: The data you select should have at least two groups to compare (e.g. treatment/control; different cell types; different organisms).
- Tip: For this project, I suggest using sequence-based data, as we have focused on sequence data analysis in this course, although if you wish you may feel free to use microarray data.
- Helpful links:
- https://doi.org/10.1038/s41598-020-74567-y
- https://www.bioconductor.org/packages/devel/workflows/vignettes/RNAseq123/inst/doc/limmaWorkflow.html
- https://www.bioconductor.org/packages/devel/workflows/vignettes/rnaseqGene/inst/doc/rnaseqGene.html
- https://www.bioconductor.org/packages/release/workflows/vignettes/RnaSeqGeneEdgeRQL/inst/doc/edgeRQL.html
- https://bioconductor.org/packages/release/bioc/vignettes/gage/inst/doc/RNA-seqWorkflow.pdf

8) Proteomics

- High-throughput proteomics is a fascinating discipline, which involves leveraging technologies (e.g. mass spectrometry) and genomic data to draw new conclusions about biological mechanisms, including in cell biology, protein-protein interactions, host-pathogen interactions, and drug discovery and prediction. This project will involve exploring R tools for proteomics.
- Clearly, this project would be a suitable choice for students interested in proteomics! As well, this would be a good choice for those interested in embarking on your own software tools adventure on an exciting topic not addressed in this course.
- Details: Start by running through the below recommended vignette, and aim to understand each step. Next, choose a specific project topic. Project topic 9A focuses on analytical methodology. Choose 1-2 key analytical decisions from the below vignette to vary, which could include a methods choice or a specific parameter setting. Iterate over multiple choices and explore the impact of key analytical choices upon the results. Alternatively, project 9B is biological. Find a raw proteomics dataset from the literature or a publicly available data base, and analyze the data yourself using R tools. Did you find the same conclusion as the authors?
- Helpful links:
- https://bioconductor.org/packages/release/data/experiment/vignettes/RforProteomics/inst/doc/RforProteomics.html
- (This is a detailed vignette, involving multiple packages, which is why there is just one provided link for this project; there are other packages too that you could explore, if you wish.)

9) Population Genetics

- How is genetic variability distributed within and among populations? Which genomic sites have been the targets of natural selection, leading to important adaptations? What is the population structure of agricultural species, fisheries species, or species of conservation concern?
- A project under the theme of pop gen would be a great choice for those students interested in evolutionary biology, population genetics in general, and/or bio-medical genetics research.
- Details: There are many potential specific projects under this theme. The general steps for these projects are as follows. Choose a target species of interest, for which there are suitable data publicly available. Check sample size, as typically you would want a dataset with information for at least 20 specimens from each of at least two proposed sub-populations. Data types could be diverse, e.g. DNA sequence data, microsatellite data, or genomics-derived SNP data. Possible data sources include NCBI's nucleotide database or dbSNP database or BOLD for barcode sequence data. Note that if you choose DNA sequence data, you should choose a gene that is expected to vary within species. Obtain data for your species of interest. Note that you need some way to define your proposed sub-populations; this could be based on geography or another definition of sub-population (e.g. host organism for pathogens). For sequence data, perform alignment. Depending upon what data type you have chosen, research and select a suitable F_{ST}-type metric to quantify population genetic structure. For DNA sequence data, consider performing AMOVA. Consider preparing an interesting visualization. One example would be to visualize the genotype data. Or, if your populations involve various geographic regions, you could consider making a map figure, with lines connecting regions/sites, with thicker lines representing greater population connectivity (higher F_{ST}) and thinner lines representing weak population genetic connectivity (F_{ST}). This is just one of several possible ideas for an interesting visualization for your project.
- An alternative project idea under the general theme of pop gen would be a marker choice project. For example, you could compare two different types of data (e.g. two different genes or types of markers) and see if they give the same answer or different answer in terms of population genetic structure. Does one type of data give more resolution than the other?
- Another option would be to develop a project in population genomics, such as detecting genomic areas under selection. See the PopGenome vignette, and feel free to speak with me to determine a suitable project scope if you are interested in that area.
- Relevant links:
- https://popgen.nescent.org/PopDiffSequenceData.html
- https://popgen.nescent.org/
- https://grunwaldlab.github.io/Population_Genetics_in_R/gbs_analysis.html
- https://grunwaldlab.github.io/Population_Genetics_in_R/AMOVA.html
- https://www.molecularecologist.com/2011/03/should-i-use-fst-gst-or-d-2/

10) Microbiome

- Analyzing whole microbial communities is an important area of research in multiple domains, including in human health studies, wildlife and agricultural animal studies, and environmental analysis (e.g. soil, water). A project under this theme could focus more on biology or on methodology.
- This project would be a good choice, of course, for students interested in microbiome analysis, microbiology, and biodiversity.
- Here, three different specific project ideas are briefly presented, and many others are possible.
- Methodological choices. You could build upon the DADA2 tutorial we went through together in class, using either the same data set or a different data set of your choosing. You could choose key parameters of interest (e.g. sequence filtering parameters, normalization methods) and vary them. How do these choices impact the total diversity detected (e.g. richness of species and/or Amplicon Sequence Variants)? Do these choices impact conclusions about differences between groups (e.g. sites, treatments, organisms)?
- Phylogenetic diversity. You could compare two or more groups (e.g. treatments, sites, organisms) from multiple perspectives, including ASV richness, biodiversity metrics (e.g. Shannon index, etc.), and phylogenetic diversity. Do the different measures lead to similar or different conclusions about your groups of interest? Are the cohabiting ASVs within samples significantly phylogenetically clustered (i.e. more phylogenetically related than expected by chance)?
- Functional analysis. If you wish to take your microbiome analysis in a new direction and build upon themes we discussed in class, you could consider running PICRUSt2 (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States). This is a software for functional prediction for microbial communities. Please note that PICRUSt2 is a Python script, which can be run from the command line. An example project would be to perform functional annotation of a 16S data set. What do you learn about the functions present in two different groups, treatments, etc.? How confident are the functional predictions made for your study system? (Note that the level of confidence in function assignments may vary a lot between, for example, human microbiome data and environmental microbiome samples, e.g. soil, water, etc.).
- Tip: Given the intended scope of this assignment and topics covered in class, I suggest focusing on marker gene analysis (likely 16S) rather than on metagenomes.
- Tip: You could use a data set associated with one of the Bioconductor tutorials or could seek other data sets through the literature or NCBI's Sequence Read Archive.
- Relevant links:
- https://benjjneb.github.io/dada2/tutorial.html
- https://cran.r-project.org/web/packages/picante/vignettes/picante-intro.pdf

- https://guangchuangyu.github.io/ggtree-book/chapter-ggtree.html
- https://github.com/picrust/picrust2/wiki (updated version to use)
- http://picrust.github.io/picrust/ (original link, has some helpful information)